SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV *COURSE CODE: DSA0201*

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometry and interpret real-world visual scenes using OpenCV or MATLAB. (BL2)*

Assessment Tool Weightage: 40%

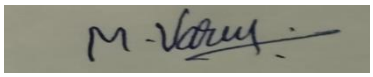
QUESTIONS: 20 *Total Marks:40*

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Vara Prasad/192324251 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Marks
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand	

Annexure A

Short Answer Test

1. A mobile app enhances photos automatically. Identify whether this task belongs to image processing or computer vision and justify.

2. A game engine generates a 3D virtual environment. Classify this under computer graphics or vision and explain.

3. A system detects edges in an image before object recognition. Which level of vision is involved?

4. A robot identifies objects after extracting features. Which vision level is this?

5. A CCTV system identifies a person from a crowd. Which vision level is involved?

6. A camera captures a blurred image due to improper focus. What aspect of image formation is affected?

7. A smartphone camera shows distorted images. What is the need for camera calibration here?

8. A self-driving car uses two cameras to estimate distance. What concept is being used?

9. A drone captures images from different angles to build a 3D model. Which concept is applied?

10. A medical imaging system analyzes MRI scans to detect tumors. Which CV application area does this belong to?

11. A virtual try-on app overlays glasses on a user's face in real time. Which fields are involved: graphics, vision, or both?

12. A system converts a real-world 3D scene into a 2D image using a camera. What process of image formation is taking place?

13. A smartphone camera app corrects geometric distortion in captured images. Which concept is applied here?

14. A surveillance system detects motion but cannot identify objects. Which level of vision is represented?

15. A robot identifies objects after analyzing their shape and features. Which level of vision is involved?

16. A VR system creates a virtual environment without using real-world images. Is this computer graphics or vision? Justify.

17. A stereo camera system generates depth maps for a scene. Which concept is being applied?
18. A system analyzes multiple images of a building to estimate its structure. Which concept is used here?

19. A biometric system uses facial features to verify identity. Which application domain of computer vision does this belong to?

20. A camera fails to measure accurate depth due to improper alignment of lenses. Which concept needs correction?

Answers

1. Image Processing or Computer Vision

Answer: Image Processing

Image processing is used to improve or modify the quality of an image. In a mobile app that automatically enhances photos, operations such as brightness adjustment, contrast enhancement, noise removal, and sharpening are performed. The main purpose is to improve the appearance of the image rather than understand its contents.

2. Computer Graphics or Computer Vision

Answer: Computer Graphics

Computer graphics is the process of creating images, animations, and virtual environments using a computer. A game engine uses computer graphics to generate 3D objects, backgrounds, lighting, textures, and realistic environments for players to explore.

3. Level of Vision in Edge Detection

Answer: Low-Level Vision

Low-level vision involves processing basic visual information, such as edges, corners, colors, and intensity changes. When a system detects edges before object recognition, it identifies boundaries and changes in brightness within an image. These features help later stages recognize objects accurately.

4. Vision Level for Object Identification After Feature Extraction

Answer: Mid-Level Vision

Mid-level vision involves analyzing and combining basic image features to understand shapes, regions, and object structures. A robot uses the extracted features, such as shape, texture, and edges, to distinguish objects. This information supports further object recognition and interpretation.

5. CCTV Person Identification

Answer: High-Level Vision

High-level vision involves understanding the meaning of visual information and recognizing objects, people, and their identities. A CCTV system that identifies a particular person in a crowd uses facial recognition or other identifying features to match the person with stored information.

6. Blurred Image Due to Improper Focus

Answer: Focus and Image Formation

Image formation describes how light from an object passes through a camera lens and forms an image on the image sensor. When the camera is not focused correctly, light rays do not converge properly on the sensor, producing a blurred image. Correct focusing helps produce a sharp and clear image.

7. Need for Camera Calibration

Answer: Camera Calibration

Camera calibration determines important camera parameters, such as focal length, optical center, and lens distortion. If a smartphone camera produces distorted images, calibration helps identify and correct geometric errors. This improves the accuracy of image measurements and the representation of objects.

8. Distance Estimation Using Two Cameras

Answer: Stereo Vision

Stereo vision uses two cameras placed at different positions to estimate the depth and distance of objects. The system compares the positions of the same object in both images. The difference, called disparity, is used to calculate depth. Self-driving cars use this technique to estimate distances to vehicles, pedestrians, and obstacles.

9. Building a 3D Model Using Drone Images

Answer: Structure from Motion (SfM)

Structure from Motion is a computer vision technique that reconstructs the three-dimensional structure of a scene using multiple images taken from different viewpoints. A drone captures images of a building or landscape from different angles. The system identifies common features across the images and estimates the camera positions to create a 3D model.

10. Tumor Detection Using MRI Scans

Answer: Medical Imaging

Medical imaging is an important application of computer vision in healthcare. A system analyzes MRI scans to identify abnormal regions, detect possible tumors, and support medical diagnosis. Image segmentation and pattern recognition can help locate suspicious areas, although medical professionals must interpret the findings.

11. Virtual Try-On App for Glasses

Answer: Both Computer Graphics and Computer Vision

Computer vision detects the user's face, identifies facial landmarks, and estimates the position and orientation of the head. Computer graphics then renders and overlays a virtual pair of glasses onto the detected face. Combining both technologies allows the glasses to move naturally with the user's head in real time.

12. Converting a 3D Scene into a 2D Image

Answer: Projection

Projection is the process of representing a three-dimensional scene on a two-dimensional image plane. In a camera, light reflected from objects passes through the lens and reaches the image sensor. The camera's projection model determines how 3D points appear as 2D image coordinates. Perspective projection is commonly used in camera systems.

13. Correcting Geometric Distortion in Images

Answer: Geometric Correction

Geometric correction is the process of correcting distortions that change the shape, position, or proportions of objects in an image. A smartphone camera may produce barrel distortion or pincushion distortion due to its lens. Image correction techniques use camera calibration parameters to reduce these distortions and produce a more accurate image.

14. Motion Detection Without Object Identification

Answer: Low-Level Vision

Low-level vision focuses on basic visual information, including changes in brightness, edges, and motion. A surveillance system that detects movement between consecutive video frames but cannot identify the objects performs basic visual analysis. Object classification or high-level vision would be needed to determine whether the moving object is a person, vehicle, or animal.

15. Object Identification Using Shape and Features

Answer: Mid-Level Vision

Mid-level vision analyzes image features such as shape, texture, boundaries, and spatial relationships. A robot uses these features to distinguish objects and understand their structure. For example, it may use the shape and edges of a cup to distinguish it from a bottle. These features support object recognition and further scene understanding.

16. VR Environment Without Real-World Images

Answer: Computer Graphics

Computer graphics is used because the VR system generates a virtual environment using computer-generated models rather than analyzing real-world images. It creates 3D objects, textures, lighting, and animations to provide an immersive experience. Computer vision is not essential unless the system also processes camera images or tracks real-world objects.

17. Generating Depth Maps Using Stereo Cameras

Answer: Stereo Vision

Stereo vision estimates the depth of objects by comparing two images captured from different camera positions. The difference in corresponding pixel locations is called disparity. The system uses this disparity, along with camera calibration parameters, to calculate the distance of points from the cameras and generate a depth map.

18. Estimating a Building's Structure from Multiple Images

Answer: Structure from Motion (SfM)

Structure from Motion reconstructs the three-dimensional structure of a scene from multiple two-dimensional images. A system identifies common features in images of a building taken from different viewpoints. It estimates camera positions and the spatial locations of visible points to reconstruct the building's 3D structure.

19. Facial Features for Identity Verification

Answer: Biometrics and Facial Recognition

Facial recognition is a computer vision application used in biometric identification and verification. The system detects a face, extracts distinctive facial features, and compares them with stored reference data. It can be used in smartphone unlocking, attendance systems, and access control. Accuracy depends on image quality, lighting, and the recognition method.

20. Incorrect Depth Measurement Due to Lens Misalignment

Answer: Stereo Camera Calibration

Stereo camera calibration is required to correct errors in the alignment and geometric relationship between two cameras. If the cameras are improperly aligned, the measured disparity may be inaccurate, leading to incorrect depth estimates. Calibration determines the cameras' internal parameters and relative positions, improving the accuracy of depth measurement.